Pre-Lab 5: Forced vibration

Submit one Word/PDF on Canvas before your own Lab 5 section. Include worked answers A-E and three figures. Use the supplied Colab notebook; no extra programming or measured lab data are required. These are illustrative parameters.

Run order and exact change

Do A by hand, then run Cell A to check. Run Cell B with J_test = 1.0, predict the effect of doubling, then change **only J_test to 2.0** and rerun Cell B. Run C and D unchanged. Answer E in your document. Keep all other parameters fixed. Run all alone does not complete the required change or explanations.

A. Predict the frequencies

For m = 1 kg, c = 1 N s/m and k = 100 N/m, calculate natural angular frequency, damping ratio and damped angular frequency, then convert both frequencies to Hz. Show substitutions and units. Explain how these differ from the imposed drive frequency. Cell A provides the formulas and checks; no figure is needed.

B. Compare two impulses

Run the J_test = 1 baseline; predict what doubling J changes. Set J_test = 2 in Cell B and rerun. Submit **Figure 1: PreLab5_Impulse_Comparison.png**, showing reference J = 1 and test J = 2 N s on the same axes. Explain amplitude and frequency using the impulse response equation in the notebook. Explain why a laboratory tap is not a measured J without a force-time record.

C. Distinguish transient and steady motion

Run Cell C unchanged: $F(t) = 1 \cos(4\pi t)$ N; initial displacement 0.1 m and initial velocity zero. Submit **Figure 2: PreLab5_Transient_Steady.png**. Calculate drive frequency [Hz]. Identify full, steady and transient responses; explain why the transient decays and which frequency remains at late time.

D. Use the actual driven-end model

Run Cell D unchanged with $k_1 = 45$ N/m, $k_2 = 55$ N/m, Y = 0.01 m and the same m and c. Here the motor moves a spring endpoint, unlike the direct-force example in C. Submit **Figure 3: PreLab5_Frequency_Response.png**, plotting $20 \log_{10}(X/Y)$ versus drive angular frequency.

Calculate the low-frequency limit $X/Y = k_2/(k_1 + k_2)$ and explain why it is not 1. Include a three-entry table with the printed numerical gain, analytical gain and relative difference. Explain what agreement checks; it is not experimental validation.

E. Prepare the measurement plan

Make a short table explaining how you will obtain: total moving mass including rods and moving damper parts; each spring stiffness; endpoint peak amplitude Y; actual drive frequency; and damping c for the same mass and damper configuration.

Show the conversions **0.024 m peak-to-peak to Y [m]** and **72 rpm to Hz and rad/s**, assuming one cycle per revolution. Explain why a change in magnets needs a matching damping estimate, and give one way to recognize unsettled motion. Bring your spring IDs and applicable Lab 4 calibration results.

Three figures to download from Colab

1. PreLab5_Impulse_Comparison.png, after J_test = 2.
2. PreLab5_Transient_Steady.png.
3. PreLab5_Frequency_Response.png.

In Colab, refresh the left Files panel, open each PNG menu and choose Download. Insert the images with readable axes, units and one-sentence captions. Rerunning cells overwrites the PNGs; download after the final required run. Do not use screenshots of the notebook interface.

Submission checklist

A: calculations and frequency explanation.
B: prediction, comparison explanation and Figure 1.
C: drive-frequency calculation, explanation and Figure 2.
D: low-frequency calculation, comparison table, explanation and Figure 3.
E: measurement plan, conversions and damping/steady-response explanation.

Use headings A-E, show your name and section, and define symbols in equations. **Submit the Word/PDF, not just the notebook or figures.** The Colab notebook contains all equations, variable definitions, per-cell instructions and code.